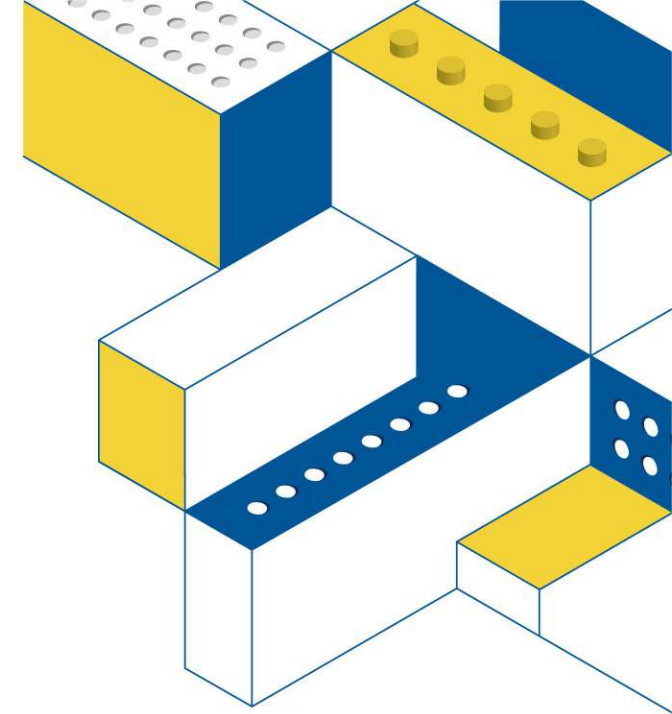


MATRIX Basic Car

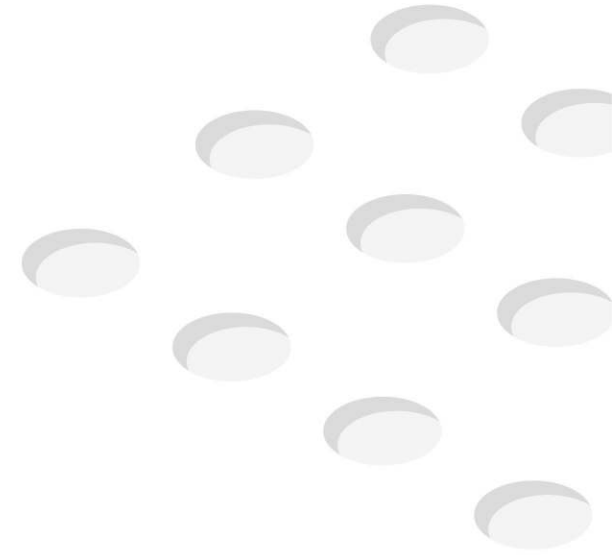
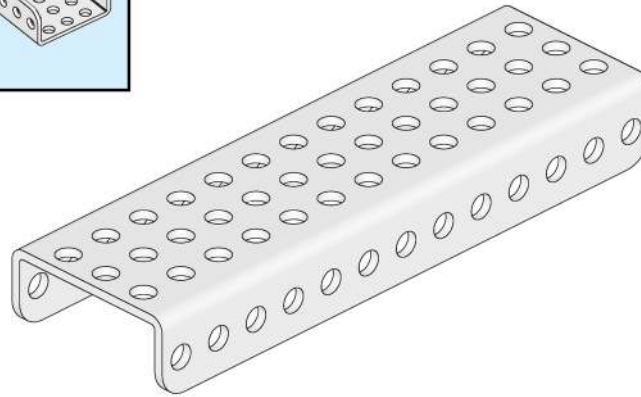
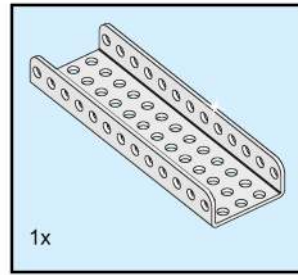
Learning objectives

- Using MATRIX components to assemble a car
- Programming for the car



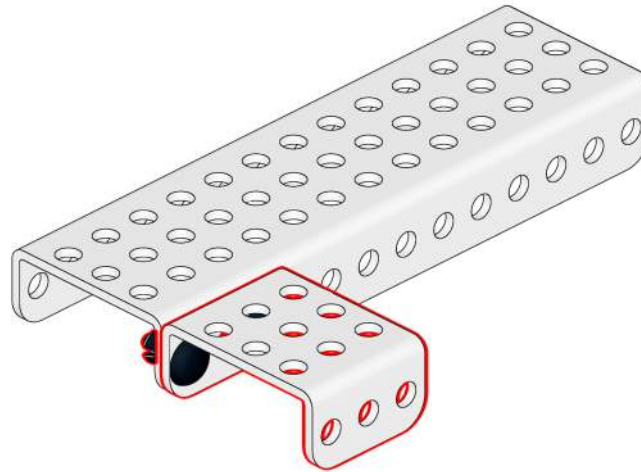
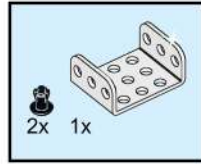
Assembly Basic Car

1



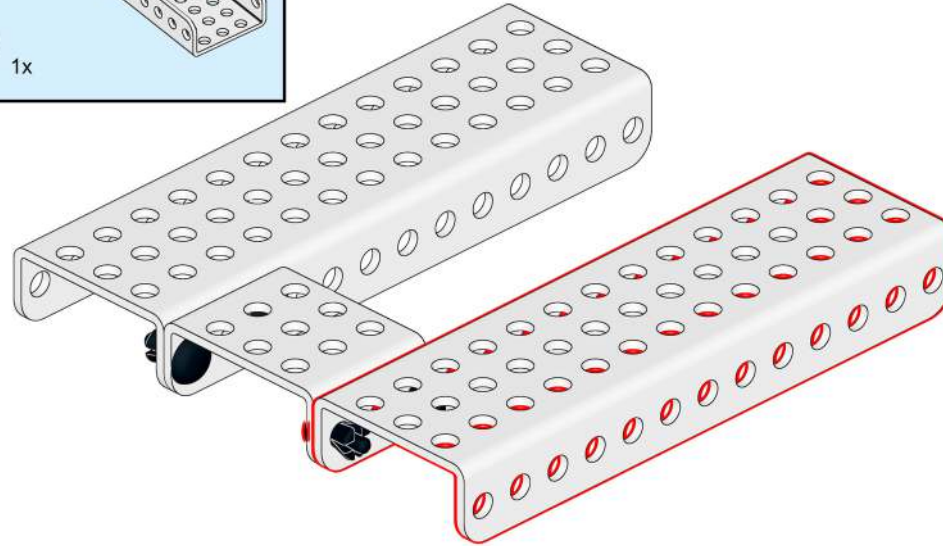
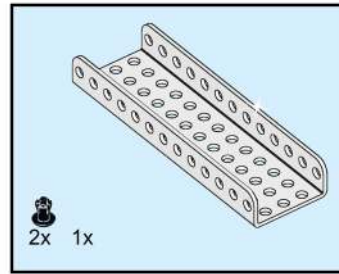
Assembly Basic Car

2



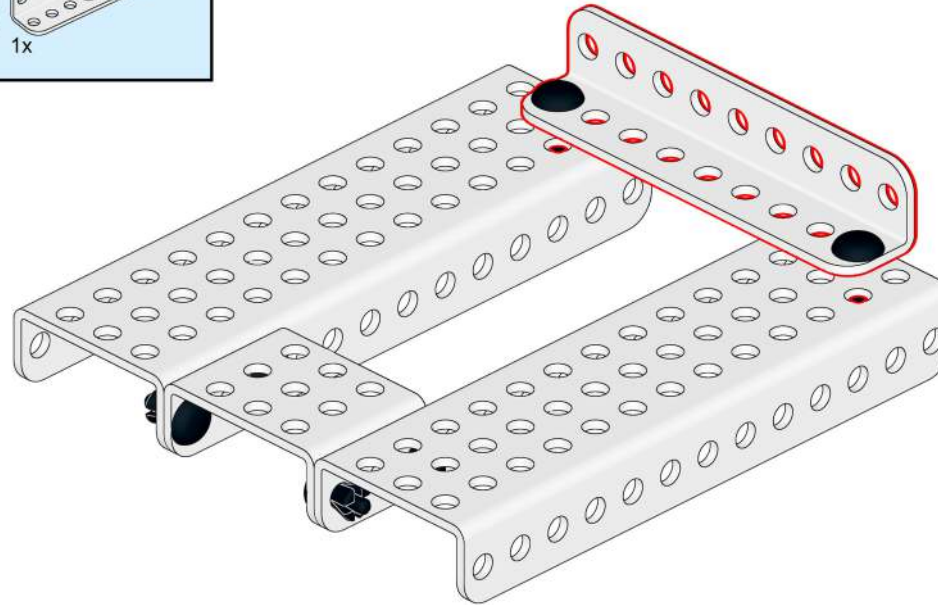
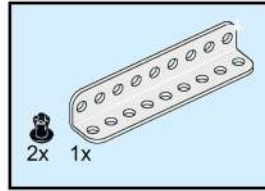
Assembly Basic Car

3



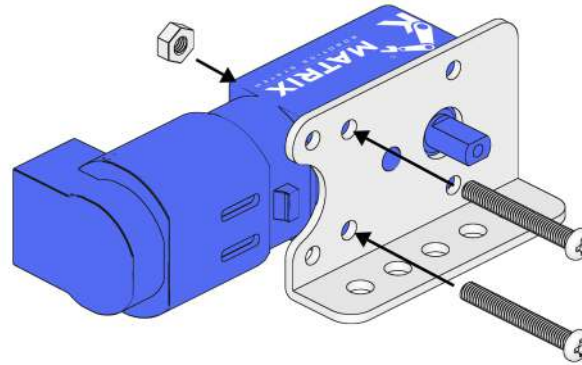
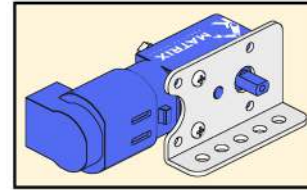
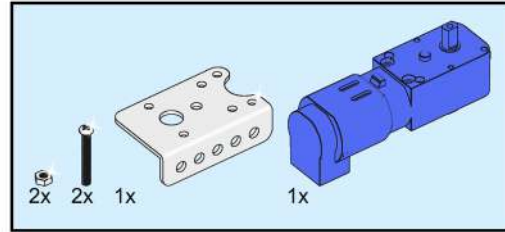
Assembly Basic Car

4



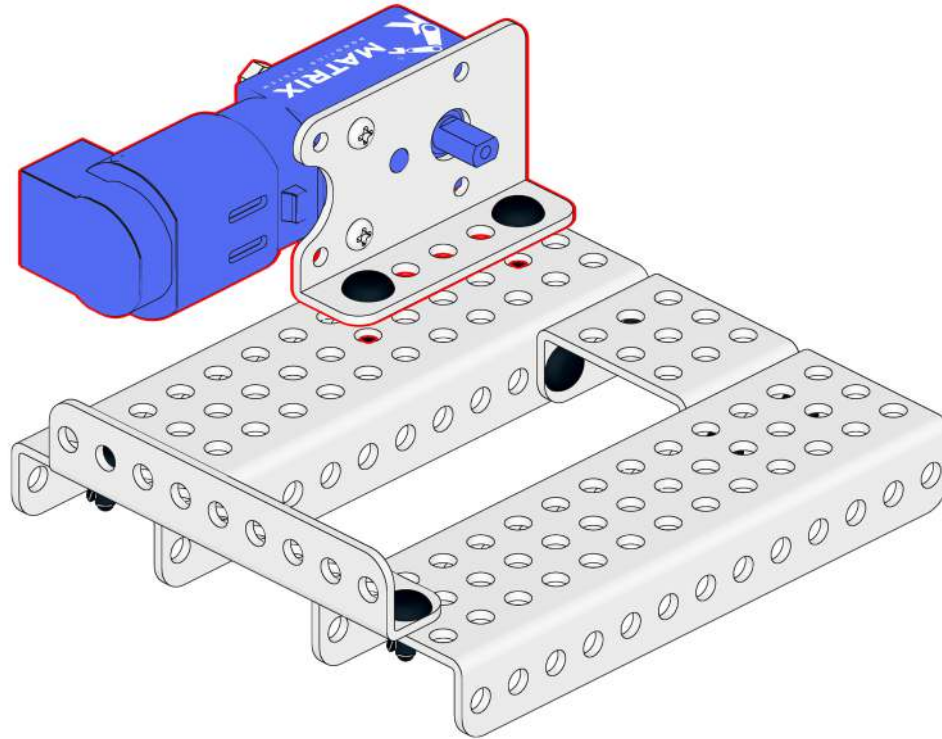
Assembly Basic Car

5



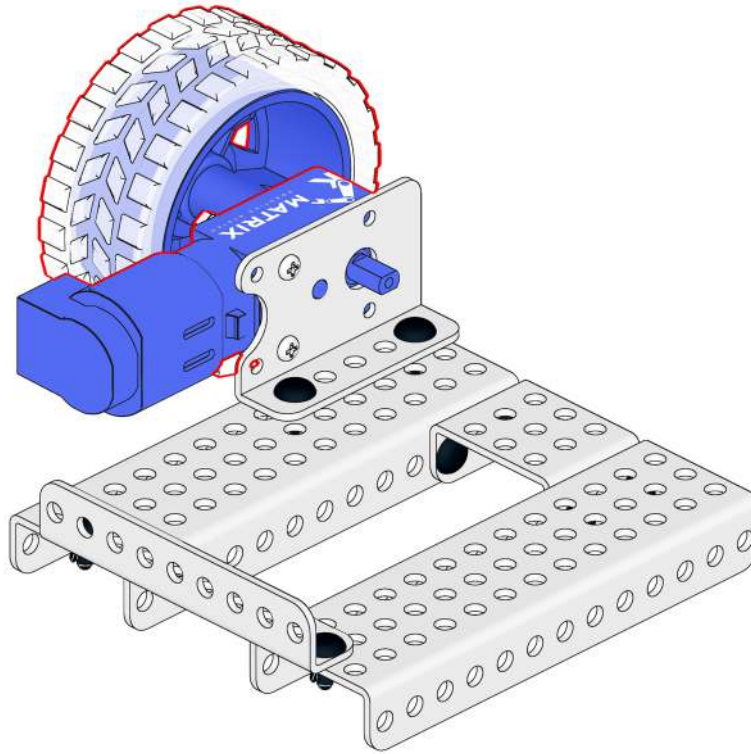
Assembly Basic Car

6



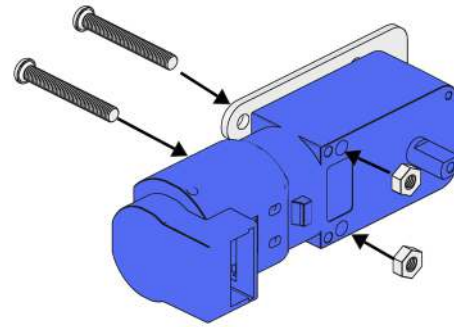
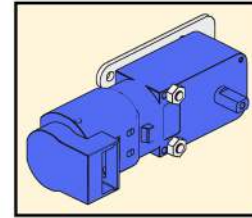
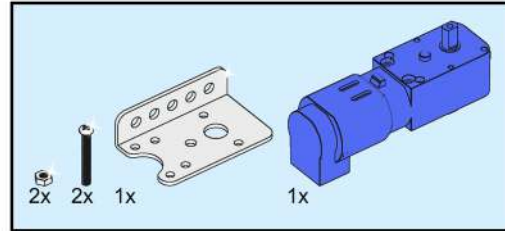
Assembly Basic Car

7



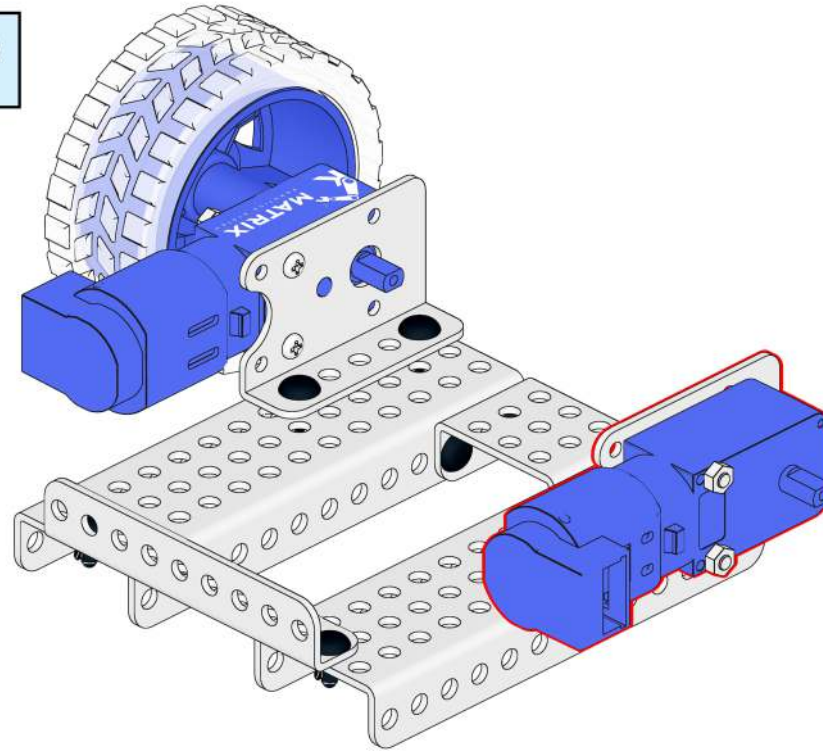
Assembly Basic Car

8



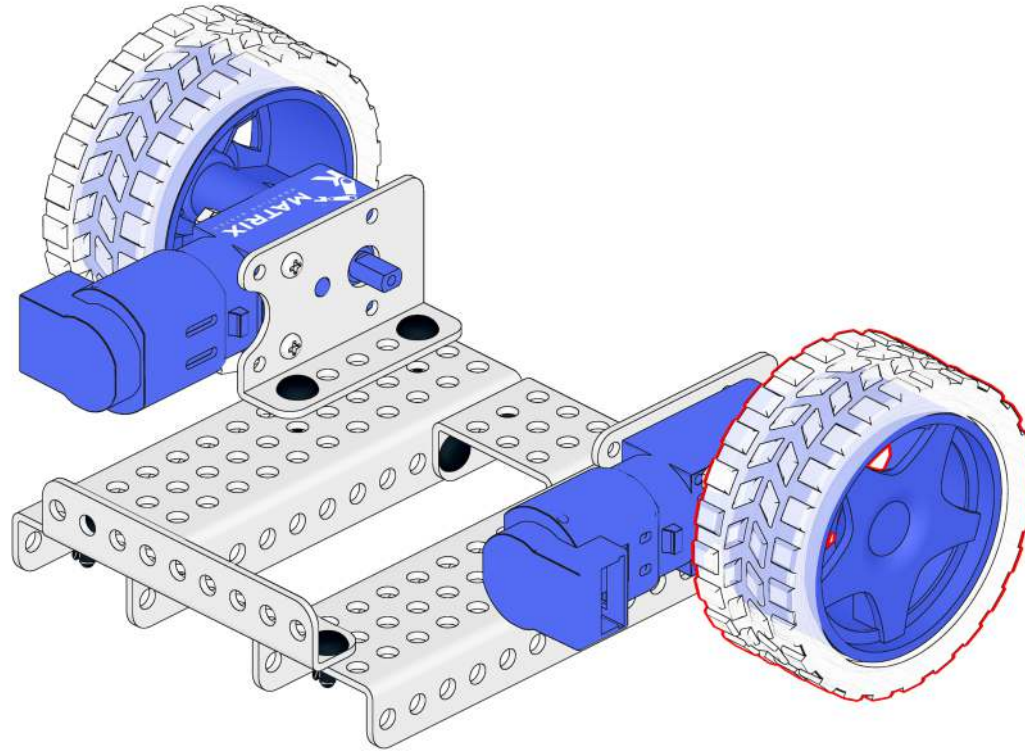
Assembly Basic Car

9



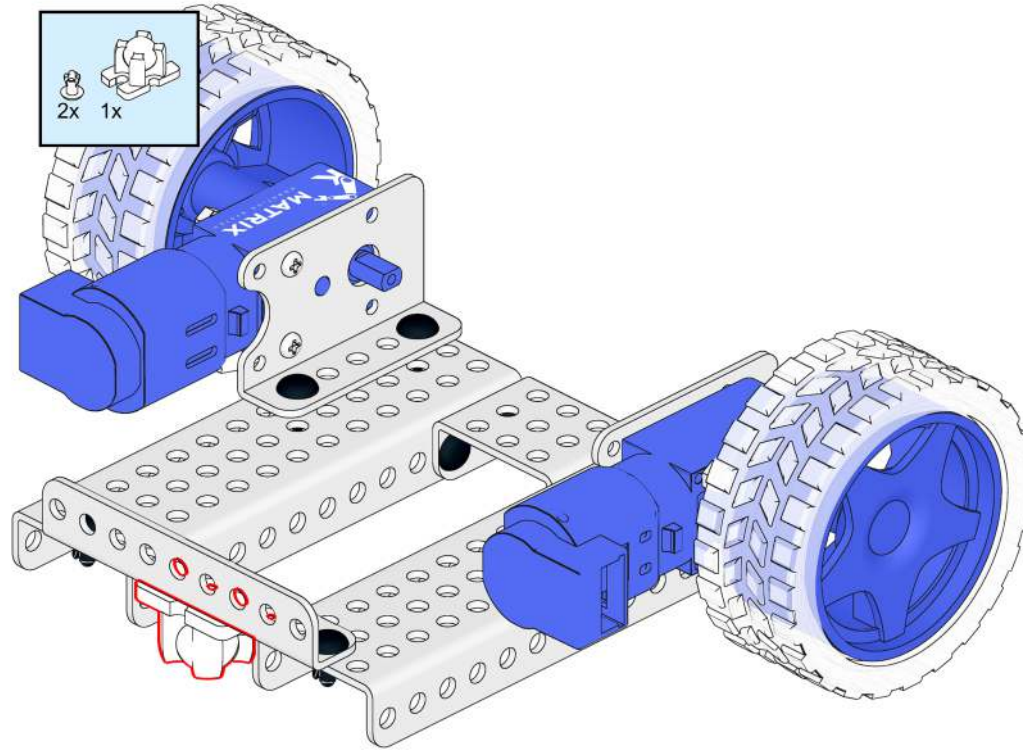
Assembly Basic Car

10



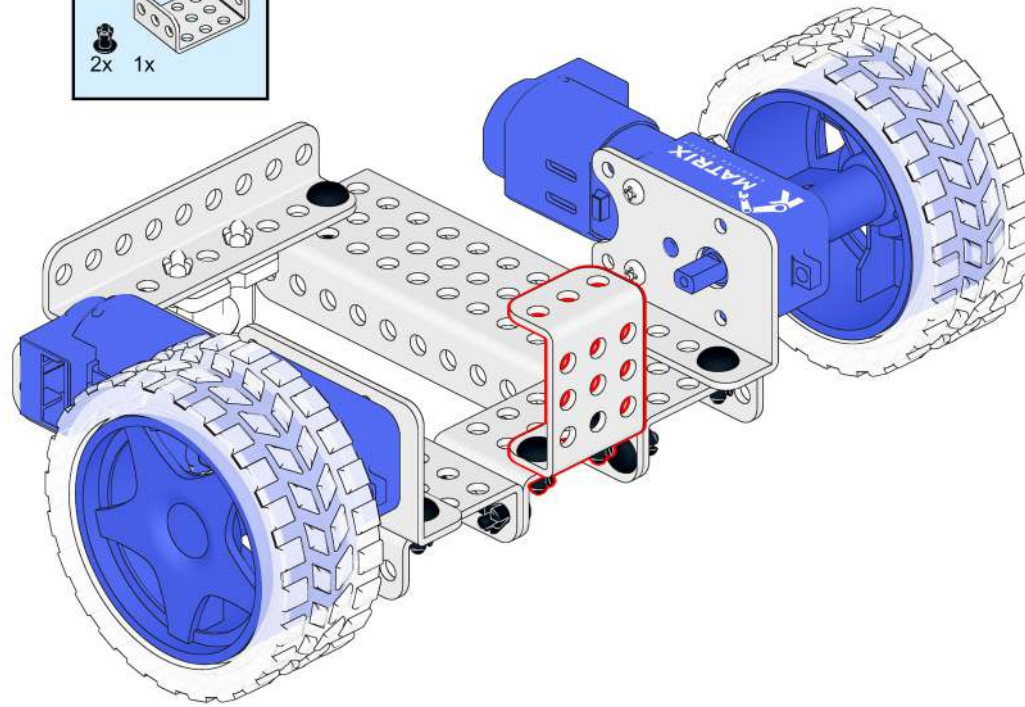
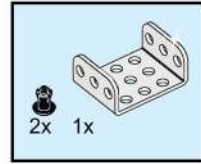
Assembly Basic Car

11



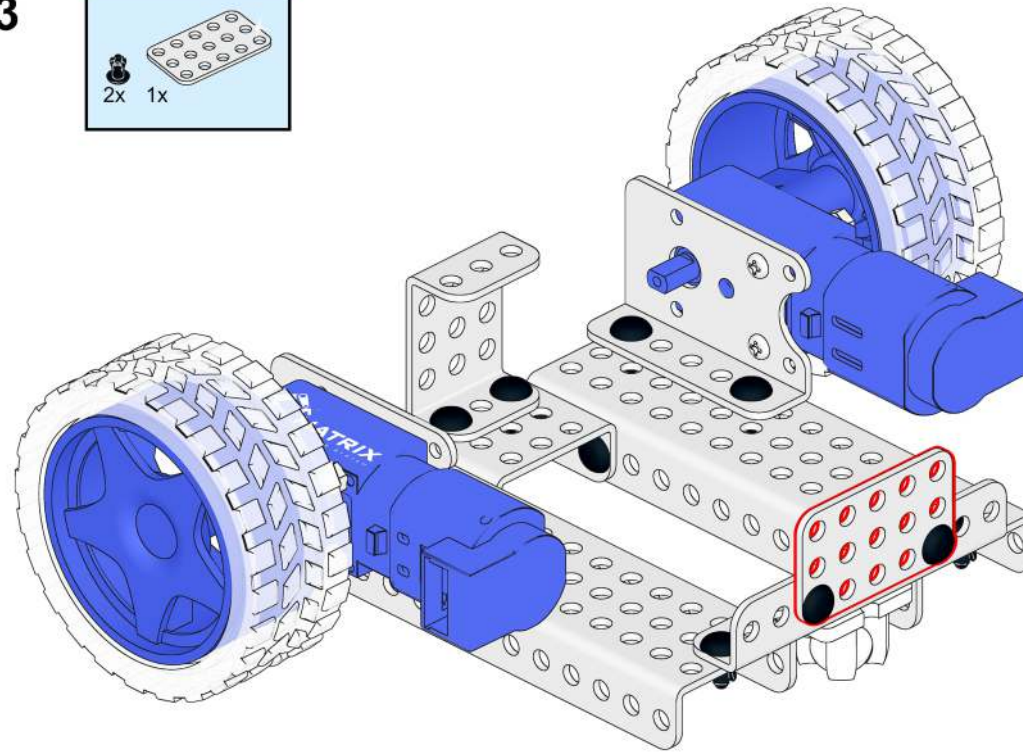
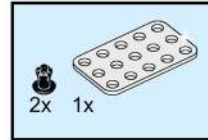
Assembly Basic Car

12



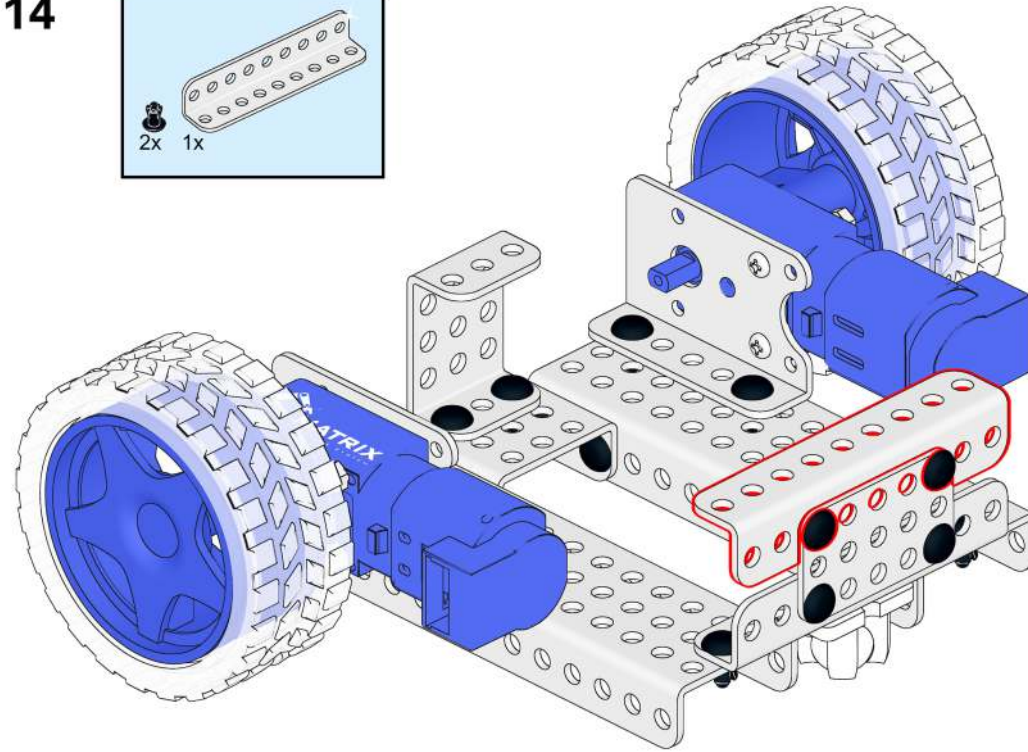
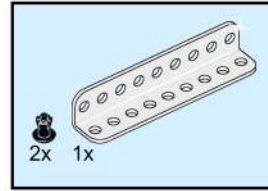
Assembly Basic Car

13



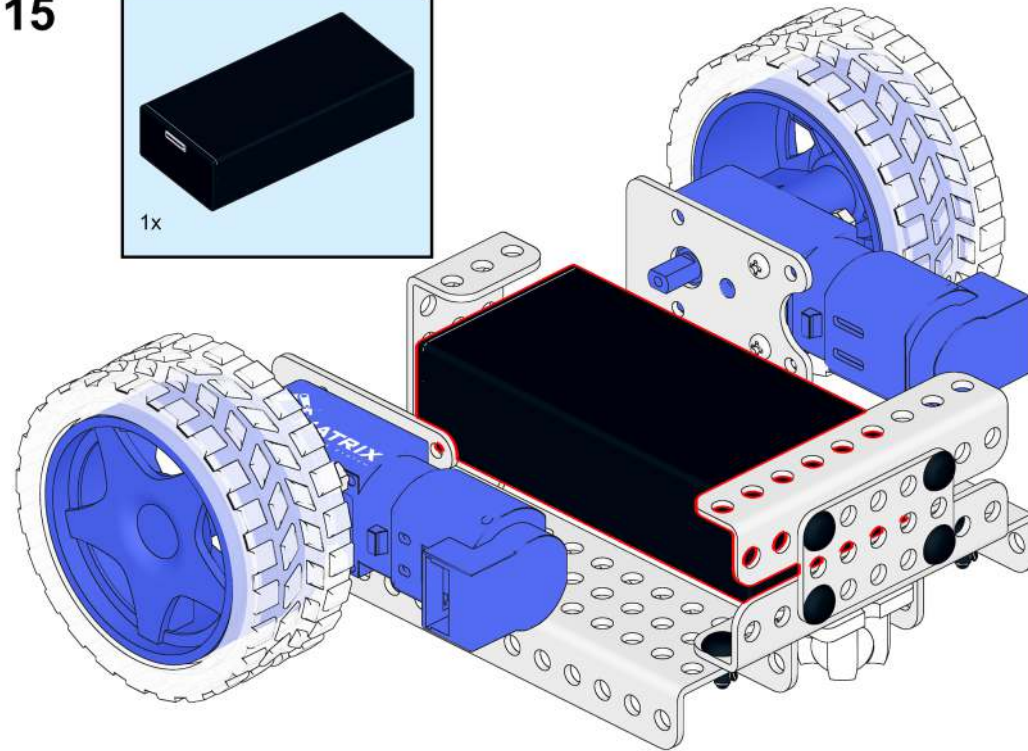
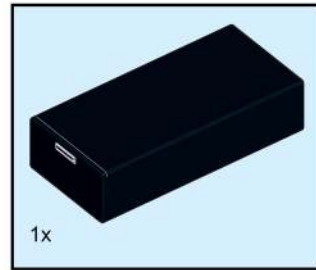
Assembly Basic Car

14

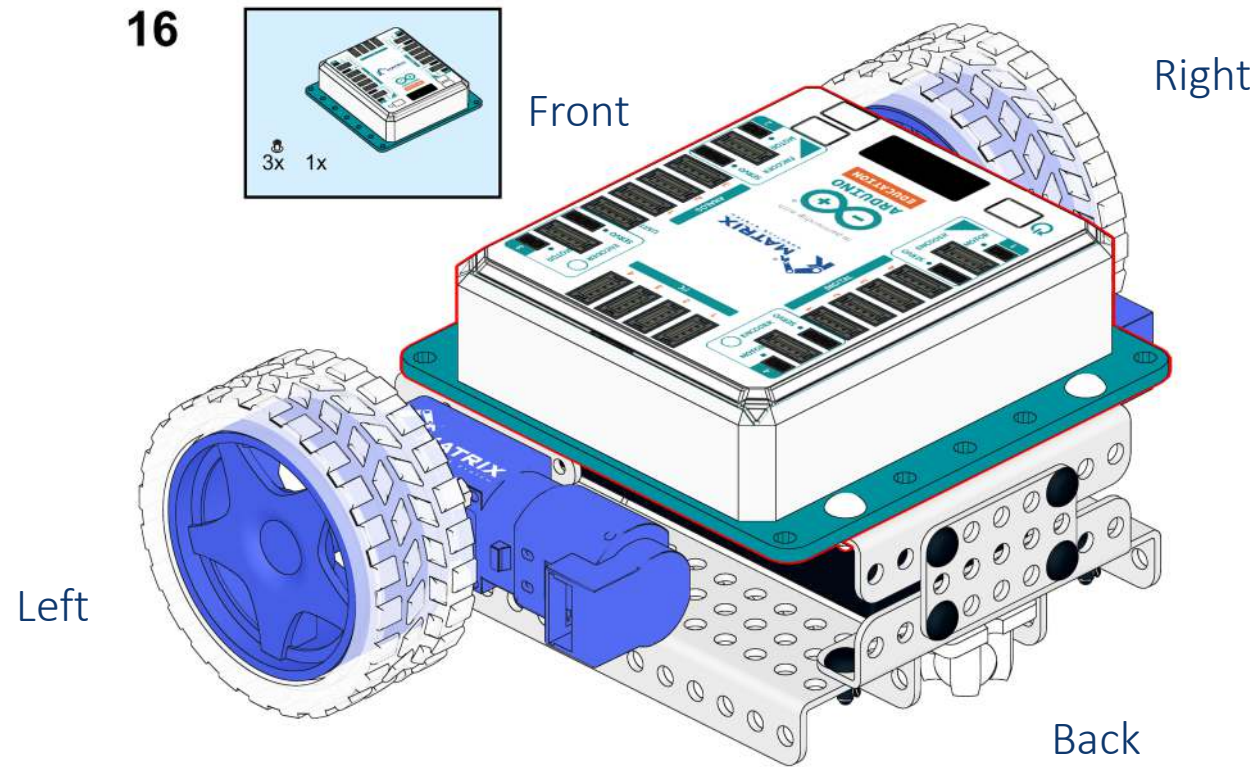


Assembly Basic Car

15



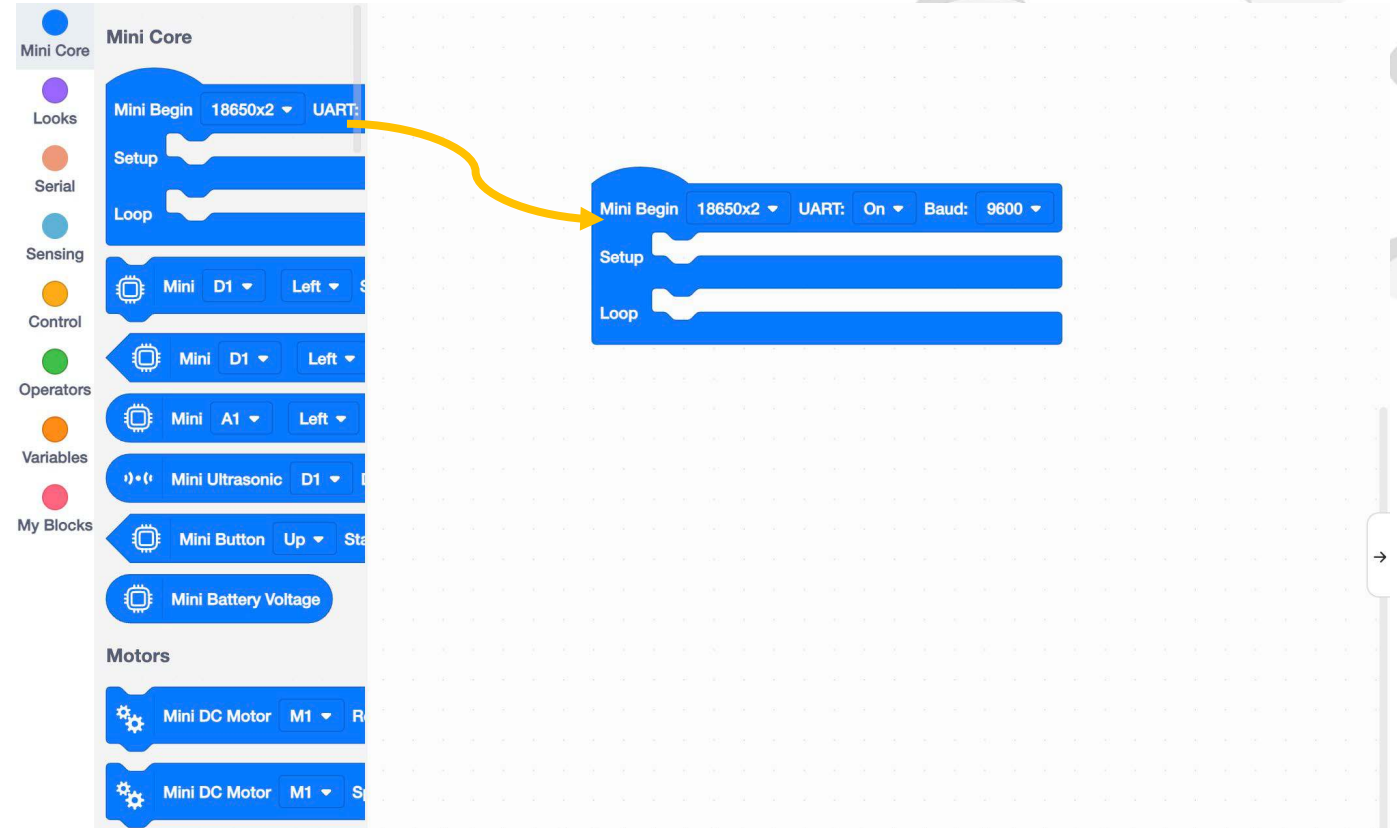
Assembly Basic Car



Left motor → Motor 1
Right motor → Motor 2
Black line near green dot "●".

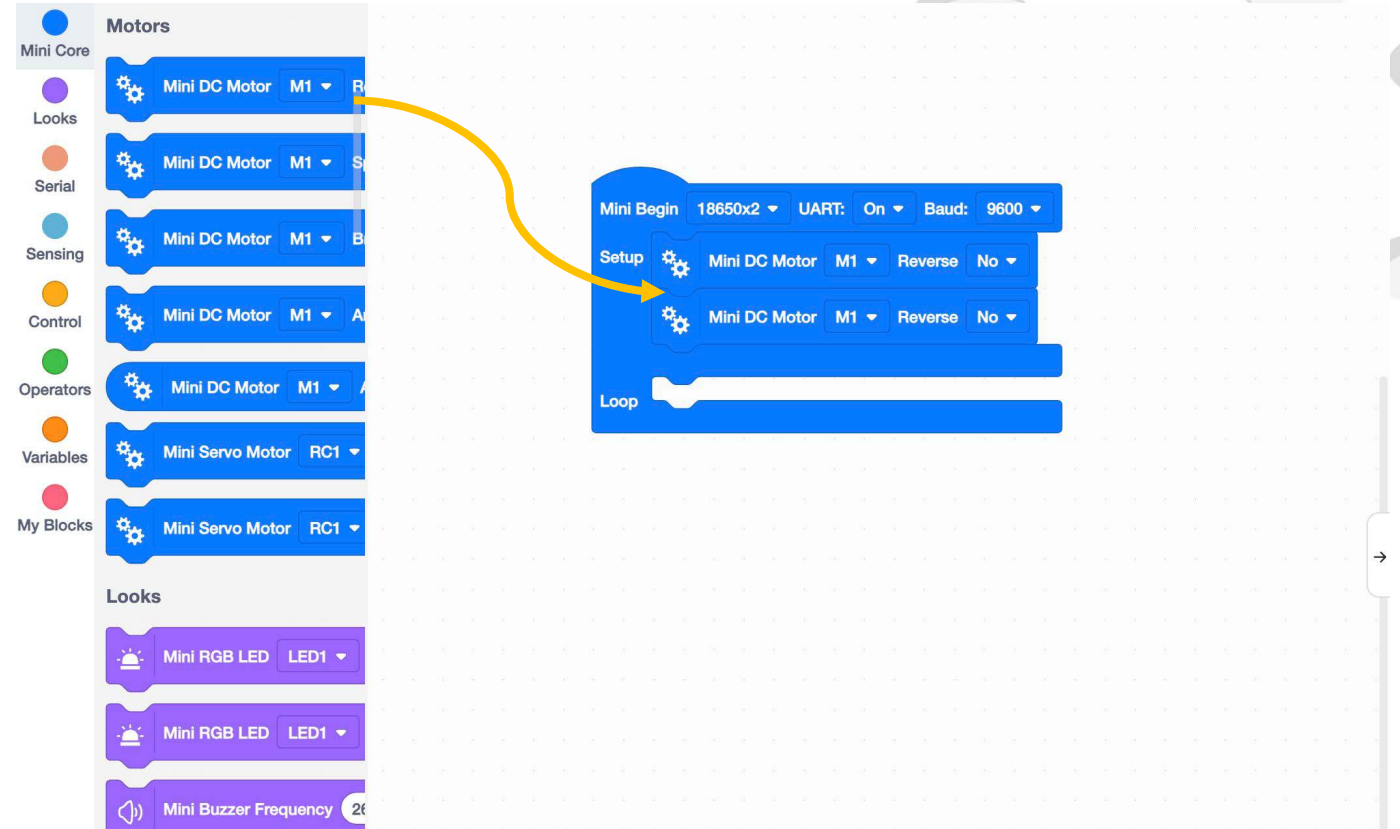
Basic Car-Go Square

1. Drag and drop a Mini Begin block from the Mini Core category into the editing area.



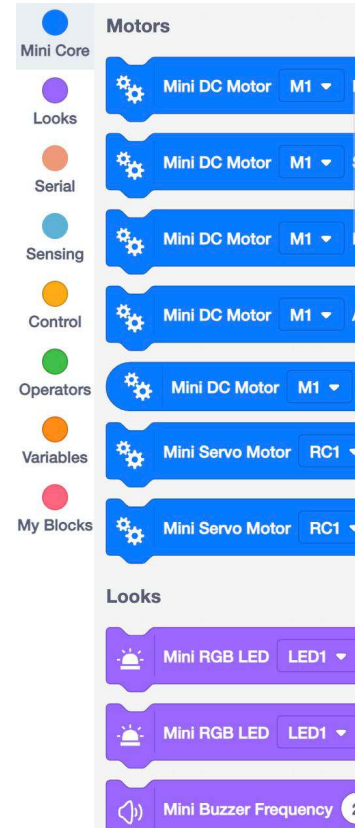
Basic Car-Go Square

2. Drag the 2 Mini DC Motor Reverse blocks to the editing area.



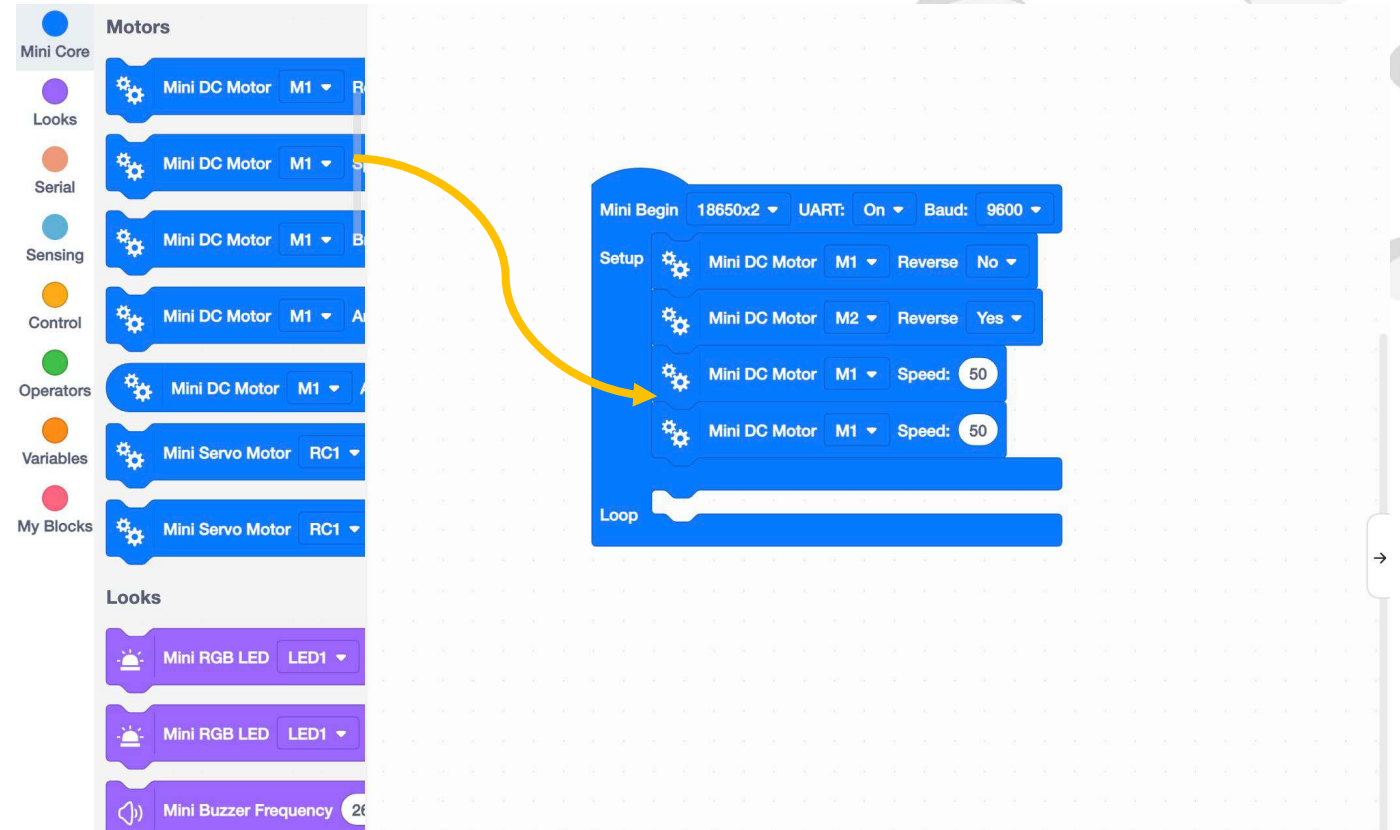
Basic Car-Go Square

3. Set the DC motor blocks to M1 and M2; because the motors are placed in different directions, one of the motors should be set to rotate in the reverse direction, and M2 should be set to rotate in the reverse direction.



Basic Car-Go Square

4. Drag 2 Mini DC Motor blocks to the editing area.



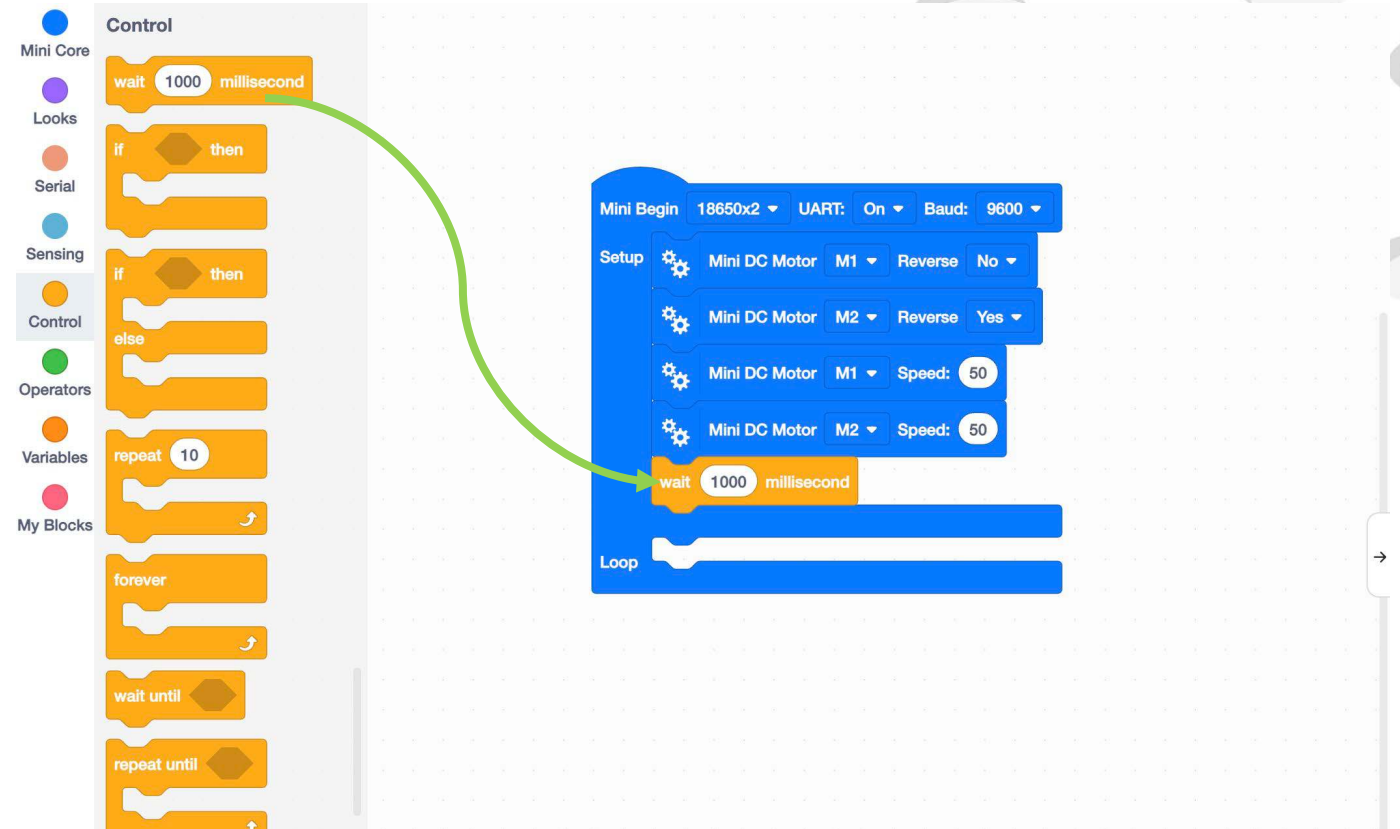
Basic Car-Go Square

5. Set the DC motor speed blocks to M1 and M2.

The image displays a block editor interface for a Mini Core microcontroller. On the left, a sidebar lists various block categories: Mini Core (blue), Looks (purple), Serial (orange), Sensing (teal), Control (yellow), Operators (green), Variables (orange), and My Blocks (pink). The 'Motors' category is selected, showing a list of blocks: Mini DC Motor (M1), Mini DC Motor (M1), Mini DC Motor (M1), Mini DC Motor (M1), Mini DC Motor (M1), Mini Servo Motor (RC1), and Mini Servo Motor (RC1). The 'Looks' category is also visible, showing Mini RGB LED (LED1) and Mini Buzzer Frequency (2000). The main workspace shows a script starting with 'Mini Begin' (18650x2, UART: On, Baud: 9600) followed by a 'Setup' loop containing four 'Mini DC Motor' blocks. The first two blocks are for M1 and M2, both set to 'Reverse' and 'No'. The next two blocks are for M1 and M2, both set to 'Speed: 50'. The 'M1' dropdown in the third block is highlighted with a red box. The script ends with a 'Loop' block.

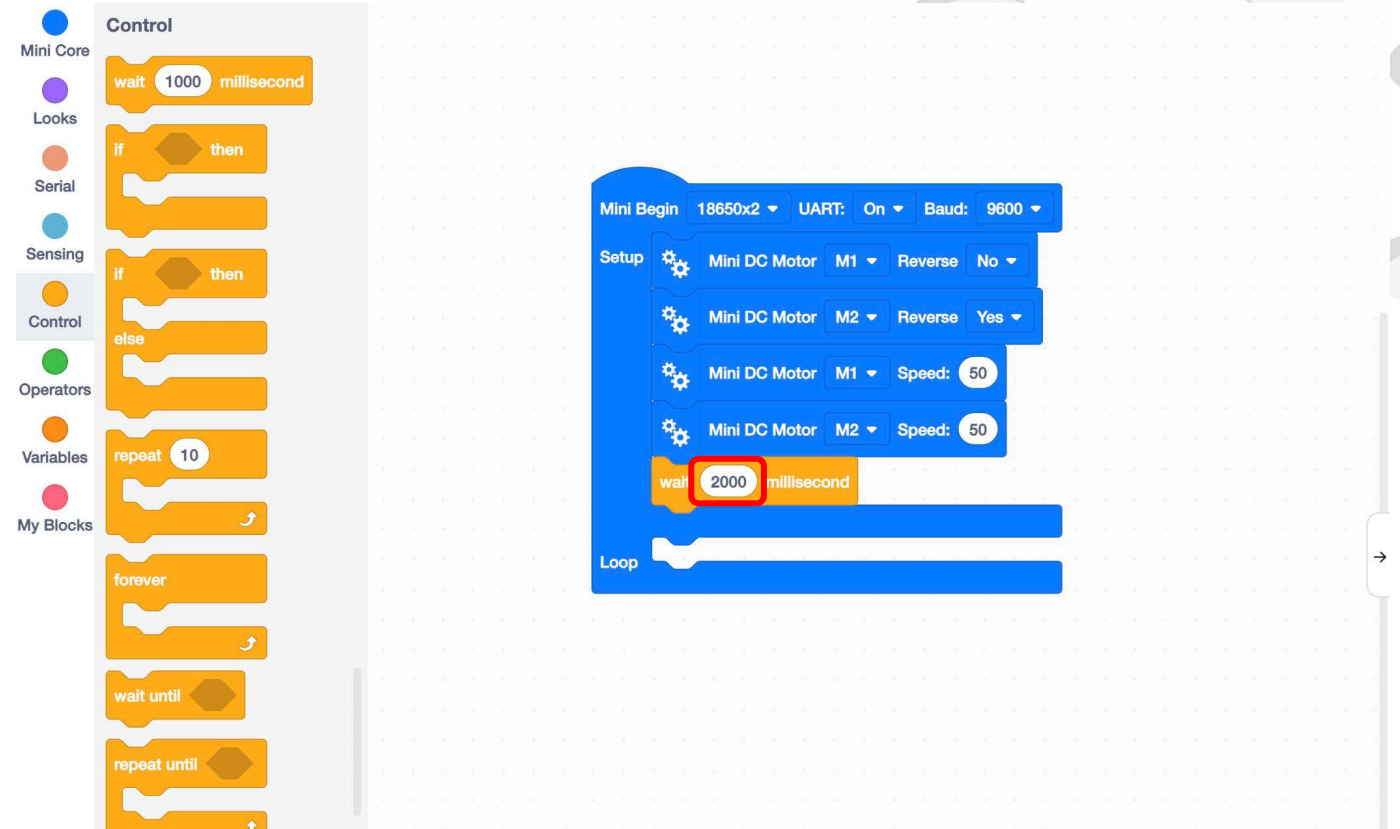
Basic Car-Go Square

6. Drag and drop a Wait ... milliseconds block from the Control category to the edit area.



Basic Car-Go Square

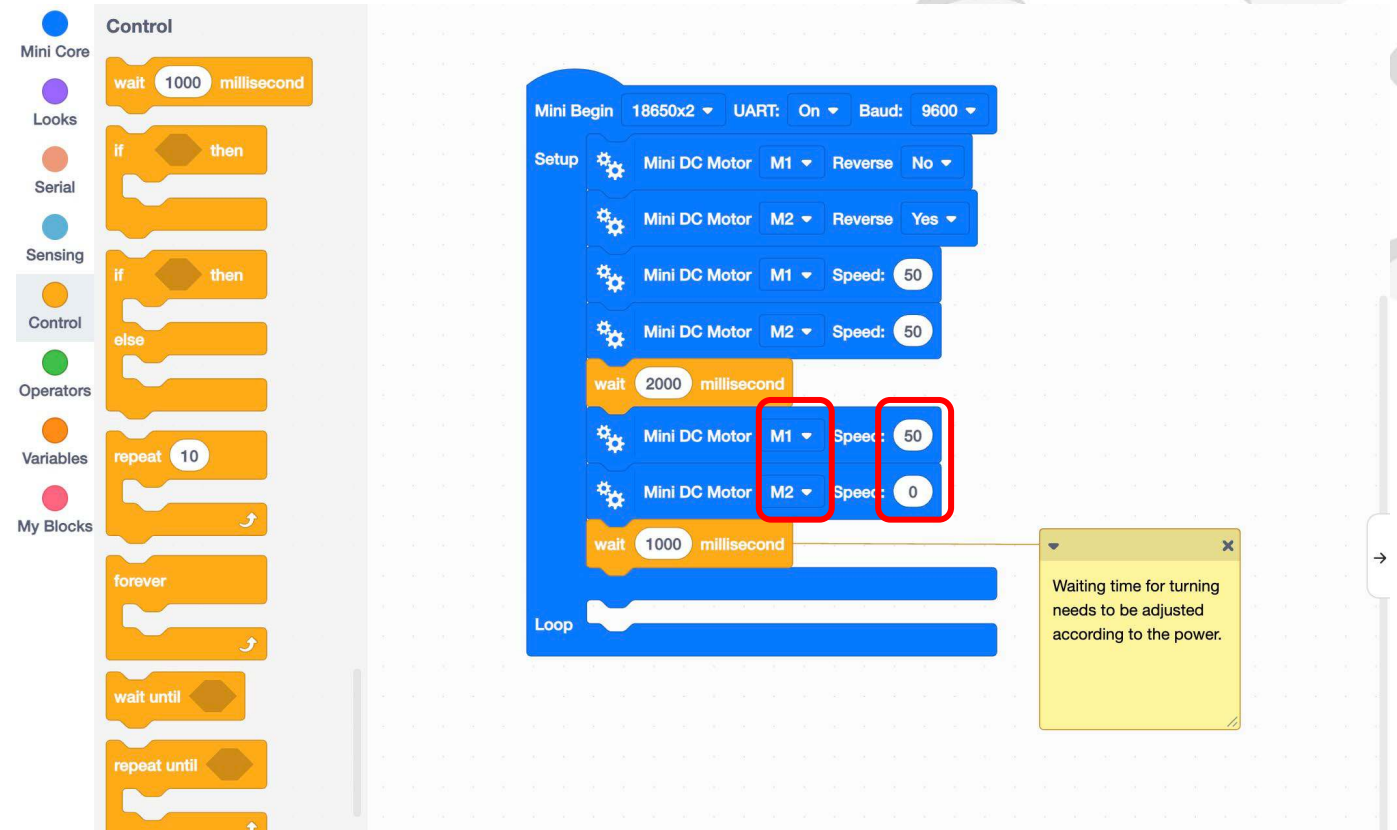
7. Adjust the time to 2000 milliseconds.(2 Seconds)



Basic Car-Go Square

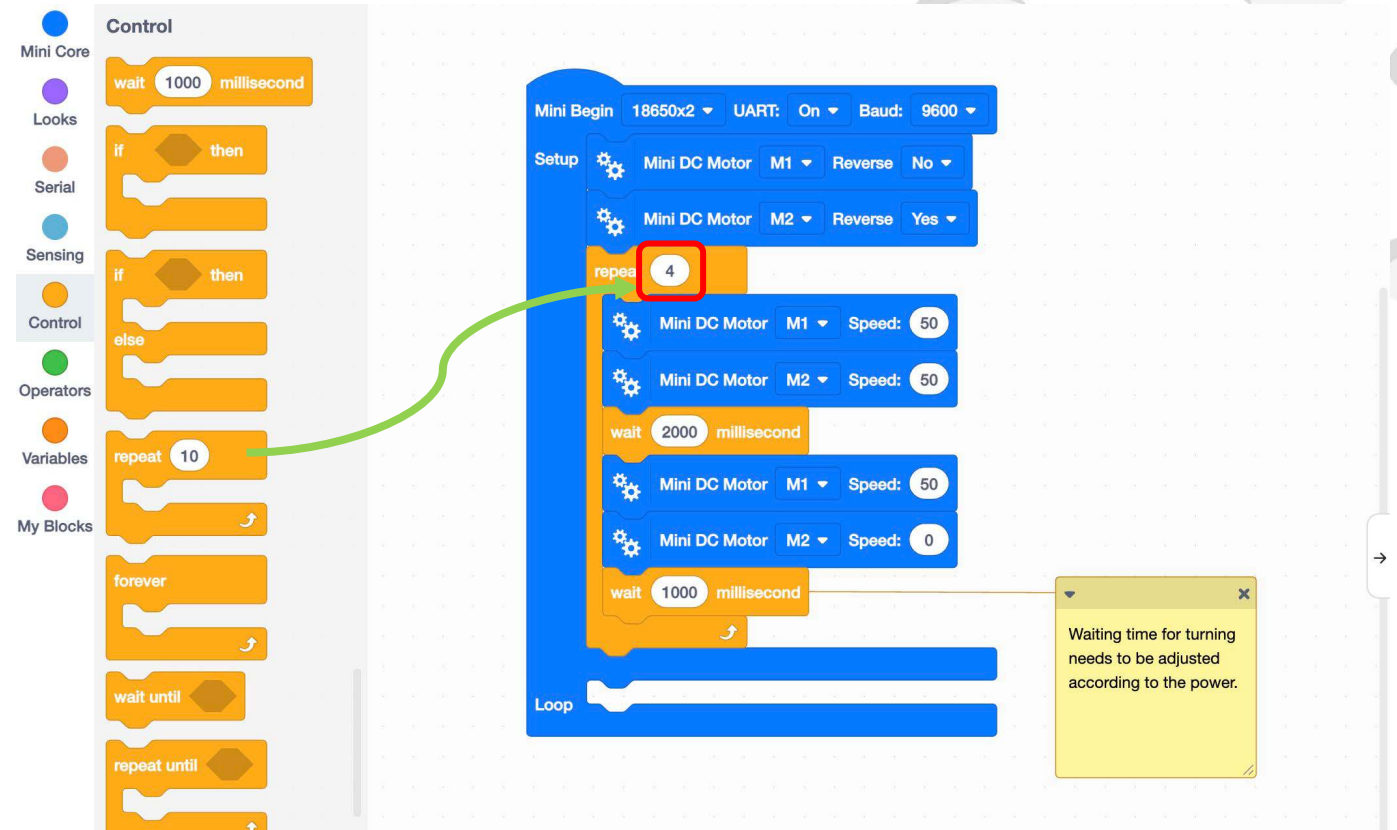
8. Add the turn program.

**** The time it takes to turn the cart 90 degrees varies depending on the friction and the power of the cart. ****



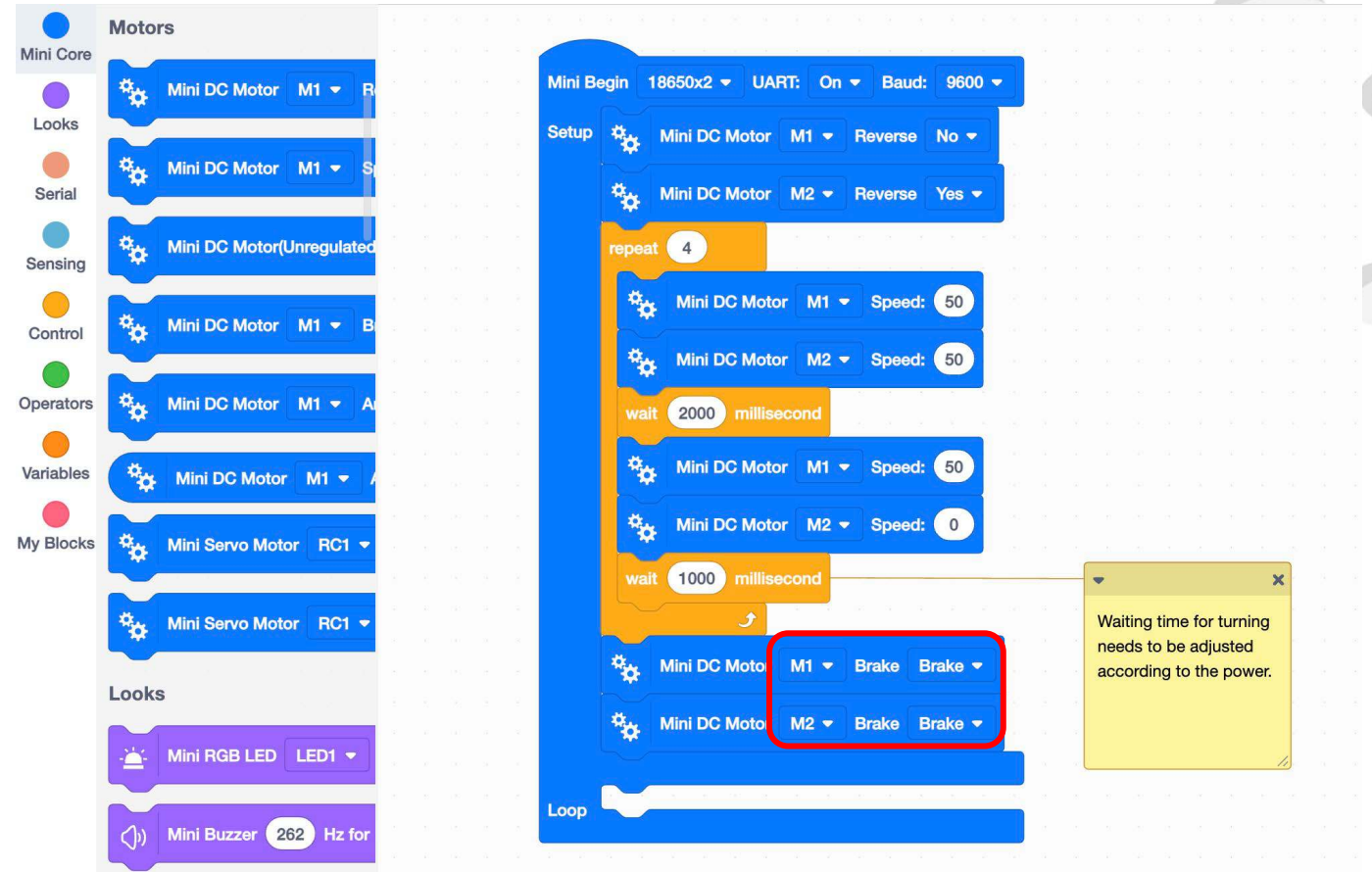
Basic Car-Go Square

9. Add the for loop and change the times of repetitions to 4.



Basic Car-Go Square

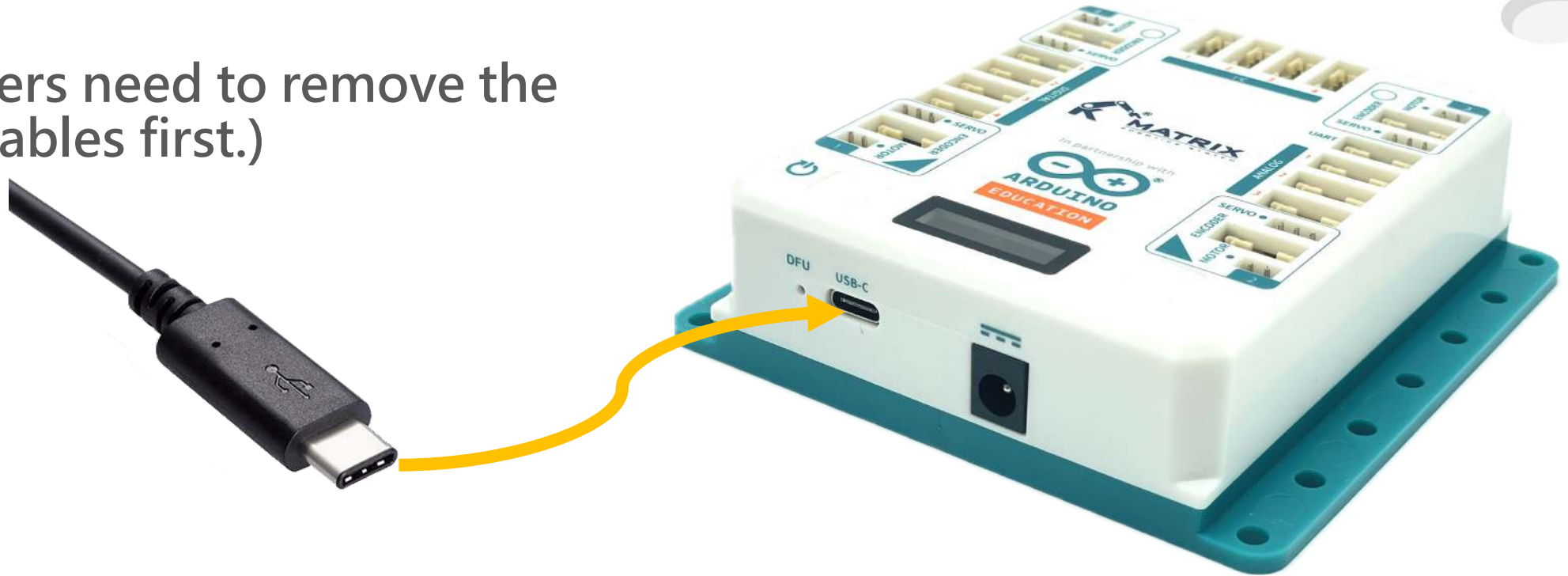
10. Set M1 & M2 brake after finishing the loop to stop the cart.



Basic Car-Go Square

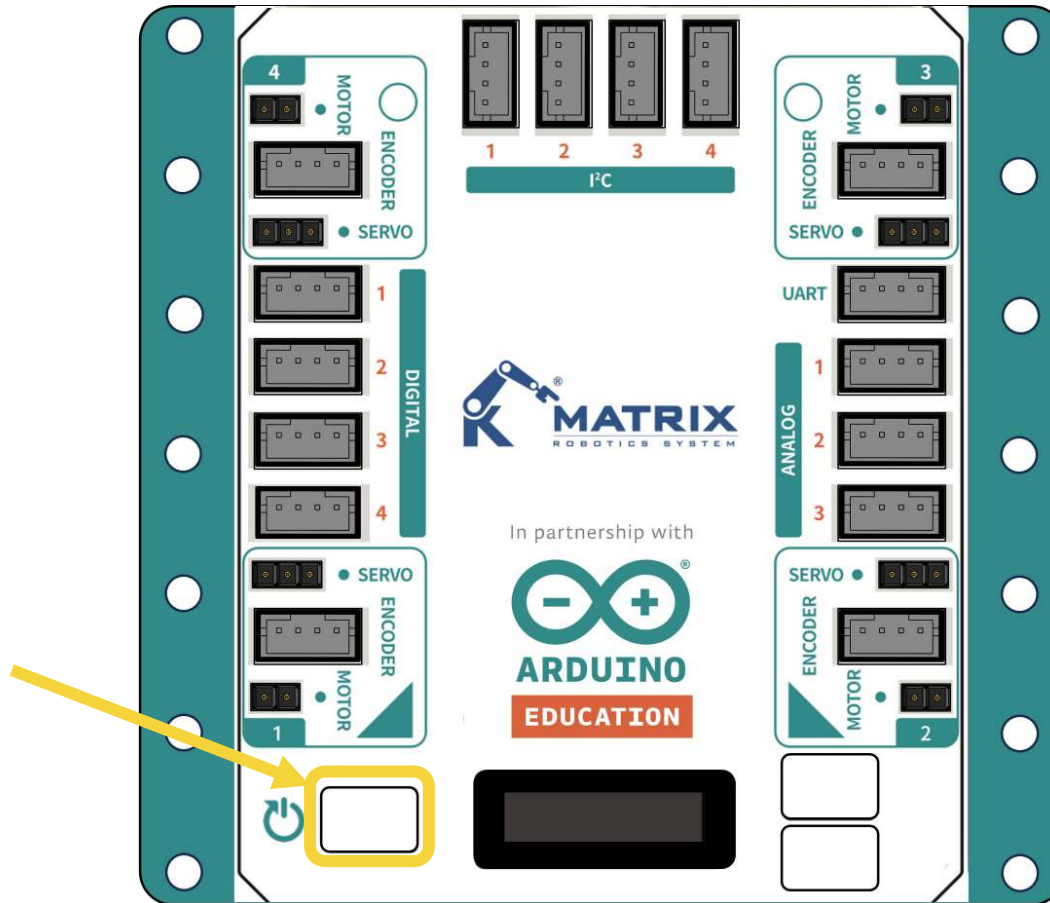
11. Connect the controller to the computer using a USB cable.

(Mac users need to remove the power cables first.)



Basic Car-Go Square

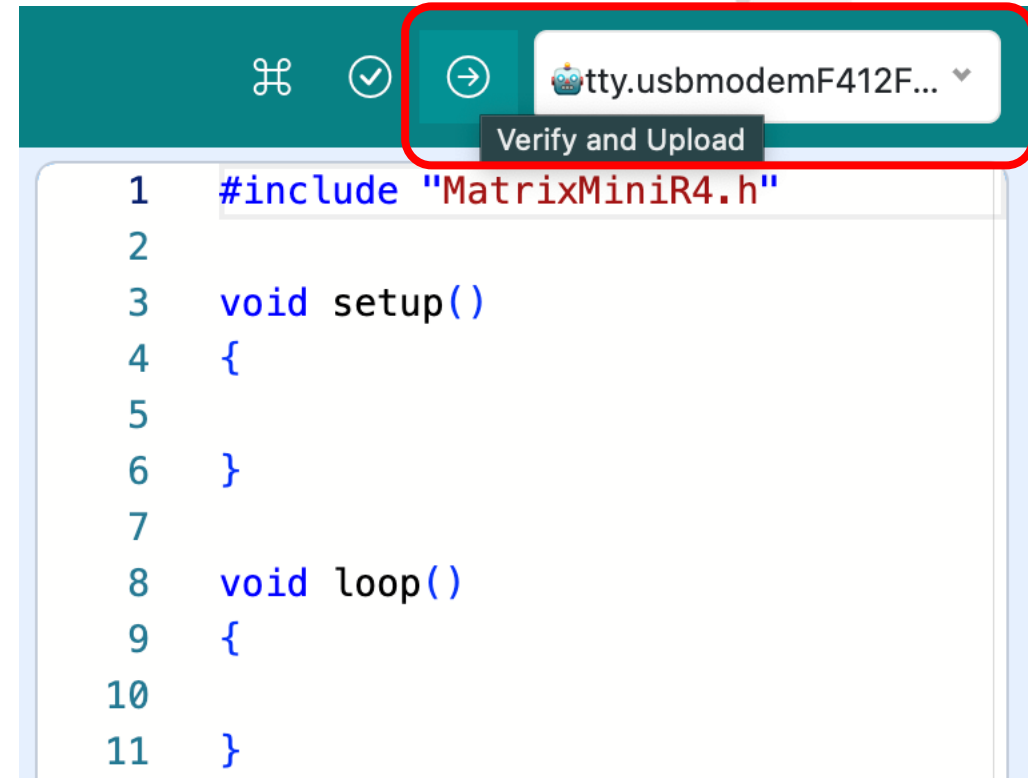
12. Press and hold the Reset Button to switch on the Mini R4.



Basic Car-Go Square

13. Choose the COM port and upload the program.

**** The R4 controller is usually the one with the robot symbol in the COM list. ****



Basic Car-Go Square

14. Connect the battery box and press the reset button to run the program.

(Make sure the battery box has switched on)





**Congratulations on
completing this unit!**

Please click on the next unit!